

**SET-5****Series %BAB%**प्रश्न-पत्र कोड
Q.P. Code **55/B/5**रोल नं.

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Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 11 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 12 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 11 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 12 questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



भौतिक विज्ञान (सैद्धान्तिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

**PHYSICS (Theory)****(FOR VISUALLY IMPAIRED CANDIDATES ONLY)**

निर्धारित समय : 2 घण्टे

Time allowed : 2 hours

अधिकतम अंक : 35

Maximum Marks : 35



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **12** questions. **All** questions are compulsory.*
- (ii) This question paper is divided into **three** sections – **Section A, B, and C.***
- (iii) **Section A** – Questions no. **1 to 3** are of **2** marks each.*
- (iv) **Section B** – Questions no. **4 to 11** are of **3** marks each.*
- (v) **Section C** – Question no. **12** is a Case Study-Based Question of **5** marks.*
- (vi) There is no overall choice in the question paper. However, internal choice has been provided in some of the questions. Attempt any one of the alternatives in such questions.*
- (vii) Use of log tables is permitted, if necessary, but use of calculator is **not** permitted.*

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$



SECTION A

1. How is the forward biasing different from the reverse biasing in a p-n junction diode ? Name one device each which works under forward and reverse biasing. 2
2. (a) Define the terms 'distance of closest approach' and 'impact parameter'. 2
OR
(b) Explain briefly any two features of photoelectric effect. 2
3. How are (a) n-type semiconductor, and (b) p-type semiconductor obtained from an intrinsic semiconductor ? Name the majority charge carriers in each of the semiconductors. 2

SECTION B

4. A hydrogen atom has only one electron yet its spectrum has many lines. Explain. Name the spectral series which lies in the visible region of electromagnetic spectrum. Mention the conditions under which the spectral line in this series will have (a) maximum, and (b) minimum wavelengths. 3
5. Briefly explain the generation and the separation of electrons and holes in a solar cell. Write two uses of solar cells. 3
6. (a) State Bohr's second postulate for hydrogen atom.
(b) The ground state energy of a hydrogen atom is -13.6 eV. Find its kinetic energy and potential energy in the ground and the second excited states. 3
7. A ray of light passes through an equilateral triangular prism and suffers minimum deviation of 30° . Calculate :
(a) the refractive index of the material of the prism.
(b) the angle at which the ray is incident on the prism. 3



8. (a) (i) Two independent monochromatic sources of light cannot constitute coherent sources. Explain.

- (ii) Two waves represented by

$$y_1 = a_1 \cos \omega t, \text{ and}$$

$$y_2 = a_2 \cos \omega t$$

produce an interference pattern in Young's double slit experiment. If the ratio of intensities at maximum and at minimum is 25 : 9, then calculate the ratio of intensities of light waves from the sources.

3

OR

- (b) (i) What is meant by total internal reflection of light ? Write the two necessary conditions for total internal reflection to occur.

- (ii) Calculate the speed of light in a medium for which the critical angle with respect to air is 45° .

3

9. A beam of light of wavelength 500 nm is incident on a photosensitive surface. If the threshold wavelength for the surface is 663 nm, then calculate :

- (a) the work function for the surface, and
(b) the maximum kinetic energy (in eV) of the photoelectrons.

3

10. Answer the following :

3

- (a) A convex lens produces a real image of an object. If the upper half of the lens is covered with a black paper, then how will the image be affected ?
(b) The time difference between the apparent sunrise and the actual sunrise is about 2 minutes. Explain.
(c) Modern telescopes use a concave mirror rather than a lens for the objective. Why ?



11. (a) (i) Write the wavelength range of the following electromagnetic radiation :

Gamma-rays, Microwaves and Ultraviolet rays

- (ii) Write one use each of the above mentioned radiation.

3

OR

- (b) (i) State Huygens' principle. How did Huygens explain the absence of secondary wavelets in the backward direction ?

- (ii) What is the shape of a wavefront emanating from (I) a point source, and (II) a linear source ?

3

SECTION C

12. When light passes through a narrow aperture or falls on an obstacle of very small dimensions, it bends into the shadow region. The bending of light waves around the corners and edges of the obstacles as well as aperture is called diffraction of light. For prominent diffraction effect, the size of the aperture or obstacle must be comparable to the wavelength of the light used. If we place a screen in front of the slit, a broad central maxima and a few alternate dark and bright bands of successively decreasing intensities on either sides of central bright band are obtained on the screen.

- (a) Light of wavelength ' λ ' falls normally on a slit of width ' a '. The diffraction due to the slit will be most prominently observed when :

(i) $a = 5\lambda$

(ii) $a = 100\lambda$

(iii) $a = 10^3\lambda$

(iv) $a = 10^5\lambda$



- (b) When monochromatic light is replaced by white light, then :
- (i) the diffraction pattern disappears.
 - (ii) the diffraction band becomes circular.
 - (iii) a white band is formed at the centre and coloured bands are formed at both its sides.
 - (iv) only a white band is formed at the centre.
- (c) If in a diffraction experiment, the width of the slit is slightly increased, then the central maximum of the diffraction pattern becomes :
- (i) broader and brighter.
 - (ii) broader and fainter.
 - (iii) narrow and brighter.
 - (iv) narrow and fainter.
- (d) If a convex lens of focal length 'f' is placed very close to the slit of width 'a' and the screen is kept at the focus of the lens, the size of the central maximum in the diffraction pattern will be :
- | | |
|------------------------------|-----------------------------|
| (i) $\frac{f \lambda}{a}$ | (ii) $\frac{2f \lambda}{a}$ |
| (iii) $\frac{f \lambda}{2a}$ | (iv) $\frac{f \lambda}{4a}$ |
- (e) A parallel beam of light of wavelength 500 nm falls on a narrow slit. The diffraction pattern is observed on a screen 1 m away. The first minimum is at a distance of 2.5 mm from the centre of the screen. The width of the slit is :
- | | |
|---------------|--------------|
| (i) 0.16 mm | (ii) 0.20 mm |
| (iii) 0.40 mm | (iv) 0.60 mm |
- $5 \times 1 = 5$